

Ibrahim Ali

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EDUCATION

- Vellore Institute of Technology** Chennai, India
B.Tech in Computer Science and Engineering; CGPA: 8.7 July 2020 – May 2024

EXPERIENCE

- Air India Limited** Kochi, India
Software Engineer August 2024 – Present
 - Backend engineer building highly available, large-scale event-driven backend systems for airline passenger communication platforms.
 - Owned and scaled a real-time high-throughput event-driven distributed delivery platform processing **~3,500+ flight events/day** and delivering **~500,000 passenger notifications/day**, replacing a polling-based scheduler that generated **~30,000 daily API calls to the Amadeus reservation system** with an event-driven model requiring only one call per event, significantly reducing external system load.
 - Optimized the notification processing architecture, reducing operational costs from **Rs.40L/month to under Rs.10L/month** through improved event processing efficiency, channel optimization, and system consolidation.
 - Migrated Air India's notification delivery platform to an **event-driven distributed microservices architecture** using **Azure Event Hub, Service Bus**, and asynchronous messaging patterns, scaling the system to handle **~1M+ notifications/day** with peak loads exceeding **50,000 requests/hour**.
 - Architected a distributed scheduling engine orchestrating time-based notification workflows for **100,000+ booking events/day** and **400,000+ scheduled notifications**, combining event-driven booking ingestion and periodic flight schedule evaluation to trigger configurable notification pipelines.
 - Implemented a configuration-driven rule engine supporting dynamic trigger offsets, flight/route targeting, passenger segmentation, and quiet-hour aware scheduling, enabling reliable notification delivery through fault-tolerant asynchronous pipelines with built-in retry and recovery mechanisms.
 - Built an **event-driven flight schedule change processing system** that ingests reservation updates, evaluates delay and prepone scenarios through rule-based validation, and automatically triggers passenger notifications, eliminating a multi-team manual workflow and reducing operational costs by **~40%**.
 - Engineered a scalable survey distribution pipeline processing **150,000+ burst records** through **Azure Event Hub**, introducing **Azure Data Factory**-based orchestration to regulate third-party API throughput and **Service Bus**-driven asynchronous callbacks to reliably process generated survey URLs and dispatch customer notifications with retryable state tracking.
 - Introduced a **WhatsApp & Email-first notification strategy**, reducing SMS and robocall costs by **~60% per month** while improving delivery reliability and customer engagement.
 - Designed and developed an **AI-assisted MCP client & server** with tool-based access to backend **Spring Boot** services, enabling secure on-demand database queries and API calls to retrieve live production system state and reducing investigation time from hours to minutes.
 - Multiple Excellence Awards (2024–2026)** — Received **Dream Team Award (1x)**, **High Flyer Award (2x)**, and **AI IMPACCT Champion Award (2x)** for driving system reliability, production stability, and operational optimization across critical passenger communication platforms.
 - Maintained production-grade reliability through structured **PR reviews**, **SonarQube analysis**, automated unit testing, and **GitHub security scanning**, while implementing **Grafana dashboards** for real-time system monitoring and streaming finalized notification records to **Databricks** for analytics and operational insights.

PROJECTS

- Flight Disruption Management System (Prototype):** — Python, React.js, Claude Code, LangChain, PSQLE
 - Designed an **AI-driven 4-agent architecture** (analysis, severity scoring, option generation, decisioning) for disruption impact evaluation and resolution generation.
 - Independently enhanced and production-hardened the system by introducing **deterministic state transitions** across **5 conversation stages**, improving reliability and traceability.
 - Expanded agent coordination logic and memory handling, reducing manual intervention during disruption spikes and improving response consistency.

SKILLS

- Languages: Java, C/C++, Python, SQL, JavaScript
- Backend: Spring Boot, REST APIs, SOAP APIs, Microservices, Event-Driven Systems
- Distributed Systems: Event Streaming (Kafka, Azure Event Hub), Messaging (Service Bus)
- Cloud Platforms: Azure, AWS (S3, EC2, Lambda, CloudWatch), Cloud-native distributed architectures
- Containerization & CI/CD: Docker, Kubernetes, Azure Pipelines
- Databases: PostgreSQL, MySQL, CosmosDB, Redis
- Foundations: Object-Oriented Design, Design Patterns, Data Structures